

PAC FOC v4.4.2 Release Notes

2023-12-08

PAC FOC Firmware Overview

- The PAC FOC application firmware is designed to showcase motor control using sensorless Field Oriented Control (FOC) on PAC devices.
- The firmware package contains example FOC application source code, Windows graphical user interface (GUI), and documentation to allow the user to configure the system to work with their target motor.
- Project files for IAR EWARM, Keil MDK (μ Vision) and Eclipse are included.
- The FOC application provides a configurable set of features to meet many different application requirements.



v4.4.2 Enhancements

Enhancement	Description
Add HALL fault tolerant operation	Support HALL operation with intermittent HALL sensor signals.
Support New PAC devices	Added support for PAC52710, PAC52711, PAC52400, PAC5524A, PAC5526A and PAC5527A.
Support OC with HPDAC	Added support for Overcurrent protection with HPDAC.
Add Speed Ref based stall detection	Added the speed-reference based stall detection to simplify algorithm.
Misc bug fixes	Fixed multiple typos and building errors/warnings.



FOC Feature Description (1)

Feature	Description
Basic FOC Control	Phase current re-construction, Clarke, Park, Inverse Clark, Inverse Park, SVM PWM, Iq/I _d PID, speed PID.
Motor ID and Auto-Tuning	Allows the user to extract motor parameters such as L _d , L _q , R, K _e and load inertia. These motor parameters can in turn be used to automatically calculate the best tuning parameters for the current and speed PI controllers.
Motor Position (MPOS) Options	Sensor-less Estimator, Hall-based FOC, or QEP as motor position options.
High-speed Position and Speed Estimator	Sensor-less position and speed estimator. Minimum estimator electrical frequency: 2 Hz (ceiling fan) - 50Hz (drone).
Start In Motion (SIM)	Two methods for SIM are available. SIM based on phase voltages (SIM _v), and SIM based on phase currents (SIM _i). With SIM _i , phase voltage measurements are no longer required, which frees up additional ADC channels.
Current Sampling Configurations	3-Shunt, 1-Shunt
Control Methods	Current, Speed, Power



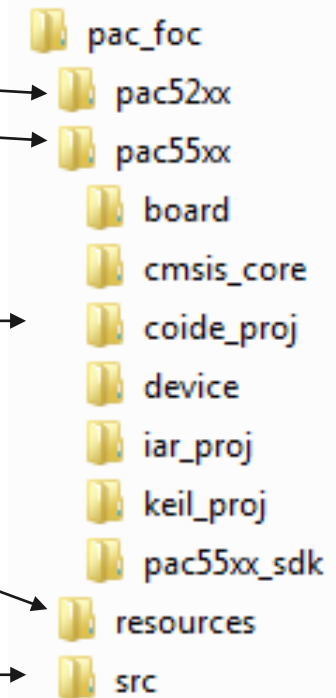
FOC Feature Description (2)

Feature	Description
Modulation Methods	SVM, Over-Modulation, SVM with harmonic injection (experimental), and generic modulations.
Performance Enhancements	Field Weakening, DQ-Decoupling, MTPA control (experimental)
Initial Position Detection (IPD)	Initial position detection algorithm that can detect with little to no audible noise.
GUI/UART	Windows based GUI configuration and control of FOC firmware over UART, for tuning of motor parameters and generation of motor parameters for insertion into firmware.
IDE Support	One set of source code supports IAR, Keil, and Eclipse
Board Configuration Files	Board configuration header and initialization files for changing configuration between existing EVKs or custom boards.
Default Parameters	Automatically generate default parameters from GUI, easy for user to tune and integrate into firmware for no GUI environment.
Debug Features	Parameter read, Debug DAC outputs, Run-time debug variable display, Control Period overflow.
Safety/Fault Features	Over-Current Protection, Open Phase Detection, Min/Max Speed Shutdown, Over/Under-Voltage Shutdown, Stall Detection.



PAC55XX Support

- In addition to supporting the PAC52XX family of devices, this FOC firmware release adds support for PAC55XX devices that incorporate an ARM Cortex M4F microcontroller with floating point support.
- The release now has separate folders for the pac52xx and pac55xx families that contain items specific to each device family.
- The device family folder contains IDE projects, SDK, board, and device files.
- The resources folder contains documentation, GUI, GUI motor parameter files, etc.
- Common source code for all devices is found in the src folder.



FOC Resources

Various resources for the FOC firmware are available in the .\resources folder of the release such as documentation, Windows GUI, GUI parameter files, etc.

- Some good documents to start with are:
 - `Guide_FOC_Getting_Started.pdf` – helps get FOC firmware up and running quickly on a PAC EVK
 - `Guide_FOC_Firmware_Overview.pdf` – describes the functionality and architecture of the firmware
- For hardware setup, consult the PAC hardware User's Guide for your specific PAC hardware.
- For motor tuning, see `Guide_FOC_Tuning.pdf`. Begin tuning a motor using the GUI by starting with the GUI parameter file called `start_foc_tuning_gui_settings.xml`.
 - execute `pac_foc_gui.exe` to start the GUI
- For other features, see the function specific guides for the various Motor Position (MPOS) options, SIM, Single Shunt, Stall Detect, etc.



Supported Tool Chains

For this release of the PAC FOC firmware the following tool chains are known to work:

Keil uVision v5.38.0

IAR Workbench v7.50.2 and v9.32.2

Eclipse with PAC v1.1.0 (Eclipse v4.27 with PAC support installer)



Thank You